卡西米尔物理的进展与展望

Advances and Prospects in Casimir Physics

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摘要

在本篇《卡西米尔效应 75 年：进展与展望》专刊的导言中，我们总结了这一物理学领域的现状，简要介绍各篇来稿的主题，提出若干尚未解决的问题，并勾勒出可能的解决路径。本文特别关注卡西米尔力的测量实验、将实验与理论对比时众所周知的传导电子耗散问题，以及新物质体系与非传统情形下的卡西米尔效应。我们的结论是：未来，这一多学科的量子效应仍将在基础物理及其应用中扮演关键角色。

1. 引言

卡西米尔效应[1]发现于 75 年前。现在正是总结它在各种物理现象中的作用、迄今所获成果与未解问题，并勾勒可能解决路径的恰当时机。

乍看之下，卡西米尔关于两块平行的、不带电的理想金属板在零温度下相互吸引的预言，其物理意义似乎相当有限——因为该力只有在极短的板间距下才显著，而"理想金属"与"零温度"这两种理想化在物理实验中都无法真正实现。

卡西米尔发现的重要性远远超出了上述预期。卡西米尔力由电磁场及其他量子场的真空涨落决定；这些涨落是一切可能有卡西米尔力参与其中的物理现象的内在组成部分。在发现之后数年，卡西米尔效应就被推广到非零温度的理想金属板[2–4]，并在 Lifshitz 理论[5–7]的框架内推广到由任意材料构成的两块厚板。人们证明：卡西米尔力是范德瓦尔斯力的推广，适用于相对论效应开始起作用的一切间距与任意温度。这类力同样存在于原子、分子与材料表面之间，此时被称为卡西米尔–波尔德（Casimir–Polder）力[9]。

由电磁场的零点涨落与热涨落产生的卡西米尔力，在凝聚态物理与原子物理中有大量应用。在凝聚态物理中，卡西米尔力作用于任意靠得较近的金属、介质与半导体表面之间（例如参见文献 [10–31]、综述 [32,33] 与专著 [34,35]）。在原子物理中，卡西米尔–波尔德力已在许多体系中得到计算[36–47]，并在量子反射[48–55]与玻色–爱因斯坦凝聚[56–61]现象中起主要作用（另见专著 [62,63]）。

卡西米尔力的许多测量是用原子力显微镜完成的——把尖锐针尖换成相对较大的球（见文献[32,34,64–78]），或用微机械扭转振荡器完成[32,34,79–85]。基于这些测量结果，卡西米尔力已被用于制造各类新一代微纳机电（MEMS/NEMS）器件[86–99]。卡西米尔–波尔德力则已在量子反射[49–52]与玻色–爱因斯坦凝聚[56,57,60,61]的实验中被测得。

电磁场之外的其他场（标量场、旋量场、胶子场等）的卡西米尔效应，在粒子物理中也有潜在应用，例如强子的口袋模型（见文献 [100–106] 与专著 [107]）。研究还表明，卡西米尔效应在拓扑非平凡的宇宙学模型中意义重大：在那里，识别条件起着与材料边界相同的作用，改变真空涨落的谱，从而给出非零的卡西米尔能量密度[108–113]。在多维物理中（额外空间维度在某一能标下被紧致化）也发现了类似效应[114–116]。

综上所述，卡西米尔力最初被视为一个有趣但更像玩具、略带异域色彩的算例，如今已发展成一个广阔的研究领域——人们常称之为卡西米尔物理（Casimir physics）。

2. 本期专刊聚焦的主题

本期专刊以《卡西米尔效应 75 年：进展与展望》为题，呈现了卡西米尔相关现象这一广阔研究领域中的若干方向。

Lifshitz 理论建立之初只能计算平行板构型；此后它被推广到任意形状的物体[117–126]，特别是实验上重要的"球位于平板上方"构型[127–135]。在本期专刊中，这一研究脉络由文献 [136] 代表——它把散射方法应用于计算与磁介质体的卡西米尔–波尔德相互作用；文献 [137] 综述了导数展开方法在卡西米尔物理中的应用；文献 [138] 考虑了周期结构中的卡西米尔力；文献 [139] 计算了导体圆锥的卡西米尔–波尔德力。

本期专刊特别关注卡西米尔力测量的近期进展。文献 [140] 综述了用原子力显微镜完成的最新实验，包括：测量非磁性与磁性金属光滑表面之间的法向卡西米尔力、褶皱表面之间的法向与横向卡西米尔力以及石墨烯体系中的热卡西米尔力。对石墨烯体系而言，把实验结果与理论相比较，要求发展描述石墨烯对电磁场响应的新方法——因为此前基于 Kubo 公式、二维 Drude 模型与密度–密度关联函数的半唯象方法[141–144]被证明不够用。新方法与测量数据一致，它使用具有非零能隙与化学势的石墨烯极化张量，该张量在 (2+1) 维热量子场论框架内得到[145–148]。

另有两篇实验论文分别研究：利用卡西米尔力的三层微机电系统对夹层液体光学性质的动态敏感性[149]；以及计划中的、两平行板在微米间距下卡西米尔压强的测量实验[150]。后一实验若能实现，将加强对牛顿引力定律的汤川型修正、以及对暗物质与暗能量假想组分（轴子、变色龙、对称子、环境依赖的伸缩子等）的限制[151–155]——这些也正是常通过测量卡西米尔力来加以限制的对象[156–162]。还有一篇偏实验的论文考虑了补偿介质与金属检验体之间静电相互作用的可能性[163]，其目标是解决表面patch（斑块）问题，该问题使卡西米尔力的测量复杂化[164,165]。

卡西米尔物理在过去 25 年间始终未解的主要问题，是如何在计算卡西米尔力时正确描述自由载流子。文献 [32,34,72,74,75,80,81,85] 表明：对金属检验体，用 Lifshitz 理论结合低频耗散的 Drude 模型算出的卡西米尔力被测量数据排除；若改用无耗散的 plasma 模型，理论结果与所有精密卡西米尔力测量实验一致[32,34,72,74,75,80,81,85]。对介质检验体，只有当计算中忽略自由载流子的作用（任何非零温度下真实介质中都存在自由载流子）时，理论与测量数据才一致[32,34,65,69,70,166]。

本期专刊中文献 [167] 的结果表明：问题的根源并不在于计算卡西米尔力时"是否"计入传导电子的耗散性质，而在于必须"正确地"计入它们。该文证明：在任何偏振的传播波区域与横磁（TM）倏逝波区域，用 Drude 模型计入传导电子在低频的弛豫性质，并不会导致与测量数据的矛盾；理论与实测卡西米尔力之间的矛盾，仅在用 Drude 模型描述金属对低频横电（TE）倏逝波的响应时出现。

众所周知，Drude 模型在传播波区域有大量实验证据，在横磁倏逝波区域也已被专门实验证实[168]；而在横电倏逝波区域，Drude 模型没有任何可靠的实验证实。据此，文献 [167] 得出结论：卡西米尔力的测量实验在横电倏逝波区域否定了 Drude 模型。该文还提出了一个经典电动力学领域的替代实验，可独立地证实这一重要结论[169,170]。本期专刊的文献 [171] 则提出另一种区分 Drude 模型与 plasma 模型的手段——基于测量由热涨落产生的洛伦兹力。

有趣的是，对本专刊文献 [172] 所考虑的两片石墨烯之间的卡西米尔力，理论与测量数据符合得很好。原因在于：石墨烯的电磁响应是在非零温度量子电动力学的严格基础上描述的，而没有使用 Drude 模型之类的任何唯象方法——这得益于采用具有任意能隙与化学势的石墨烯极化张量，该量在热量子场论框架内得到[145–148]。

本专刊中还有若干论文研究特定构型下的卡西米尔效应。例如文献 [173] 考虑 (2+1) 维场论中的卡西米尔能量，这与石墨烯、硅烯、锡烯、磷烯等新型二维材料的应用相关[174–177]；文献 [178] 讨论含缺

陷与边界的共形场论中的卡西米尔力；文献 [179] 给出处于侧向运动状态的各向同性导电平面所受的法向卡西米尔力；文献 [180] 在涨落电动力学框架内研究由相互作用物体的相对运动引起的卡西米尔–Lifshitz 摩擦力以及随之而来的加热；文献 [181] 则说明如何利用卡西米尔力来稳定由非均匀磁场中的排斥力托起的石墨烯片的悬浮。

另有几篇论文在相当非标准的情形下、使用一些替代方法考虑卡西米尔效应。例如文献 [182] 研究轴子电动力学中的卡西米尔效应：由于存在额外的赝标量，电场、磁感应强度、电位移与磁场之间的关系比连续介质的标准电动力学更为复杂。该主题与被称为拓扑绝缘体的新材料密切相关，具有很大的实际意义。

本专刊文献 [183] 讨论了基于半经典电动力学的一条有趣途径：其前言讨论了物理早期历史上关于分子间力的不同概念，并说明半经典方法如何给出人们熟知的、Lifshitz 形式的卡西米尔自由能公式。另一项研究 [184] 致力于推导沉积在介质衬底上的两个 Chern–Simons 边界层之间的卡西米尔压强，为此发展了基于电、磁格林函数的规范不变形式。还有两篇论文致力于动态卡西米尔效应：其一 [185] 讨论动态卡西米尔效应、黑洞与加速电子的辐射温度之间的类比；其二 [186] 研究作用在运动镜面上的非对称力，该镜面由"二维时空中 delta 函数与其导数之差"形式的势所建模。此外还有一篇论文研究均匀磁场中电子质量的单圈修正 [187]——该问题已有多位作者讨论，但结果存在分歧；鉴于辐射修正在物理本质上与卡西米尔效应相似，有必要澄清所有现存分歧。

如第 1 节所述，卡西米尔效应不仅出现在有材料边界的构型中，也出现在拓扑非平凡的时空中，因此它在引力与宇宙学、以及粒子物理的多维理论中扮演重要角色。本期专刊有四篇论文代表这一方向。众所周知，反德西特（anti-de Sitter）时空在宇宙学中意义重大。一般地，膜世界（braneworld）模型包含在体（bulk）中传播或在膜上局域的场；膜上的边界条件会对物理可观测量的期望值产生卡西米尔型贡献。文献 [188] 在"垂直于反德西特时空边界的两个平行膜"这一构型中，计算了标量场的表面应力–能量张量的真空期望值。

宇宙学的另一类重要客体是所谓的宇宙弦，即早期宇宙在宇宙学相变过程中可能产生的拓扑缺陷。文献 [189] 考虑两个宇宙弦之间由最小耦合标量场的真空涨落引起的卡西米尔相互作用，涵盖小间距与大间距两种情形，并计及弦的横向尺寸。

最后，文献 [190] 讨论了弱引力场中两平行板的卡西米尔效应，以及由杨–米尔斯（Yang–Mills）场

的卡西米尔能量密度所决定的虫洞；同一篇论文还研究了具有非平凡拓扑的欧几里得时空中的卡西米尔能量密度——该拓扑等价于施加所谓的螺旋（helix）识别条件。

3. 未来展望

如上所示，卡西米尔效应是一个广阔的研究领域，其影响几乎触及现代物理的每一个分支，许多国家的众多研究组正在理论与实验上积极研究它。过去几年里已取得若干突破性新结果。下面我们列出该领域未来最需要解决的关键问题。

尽管对金属检验体、在数微米以下的间距上，卡西米尔力已被高精度测量，但在更大的间距上、以及对于由半导体与介质材料构成的检验体，未来有望取得新的突破性实验成果。精密力测量的进展可以借助传统与新技术来推动（见本专刊文献 [140,150]），并通过补偿虚假的电力 [163–165] 来实现。所获结果有望用于制造由卡西米尔力驱动的新一代微纳机电器件（见本专刊文献 [149]，该文正沿着这一方向展开）。

可以预期，在通过 Drude 模型计入传导电子耗散后出现的实验与理论不一致问题（常被称为卡西米尔之谜，Casimir puzzle[32,34,75]）将很快得到解决。在本专刊中，该问题已被缩小为"Drude 模型在横电倏逝波区域不适用"[167]。已有研究表明[191–193]：若在这一区域唯象地修改 Drude 模型、加入空间非局域的贡献，理论预言就与测量数据一致。剩下的工作是把这种唯象处理建立在坚实的基本原理之上——正如人们已经为石墨烯的电磁响应所做的那样[77,78,145–148]。

在不远的将来，卡西米尔物理还将在超越标准模型的现代理论途径中找到新应用，例如膜模型、具有紧致化额外维度的多维物理、拓扑缺陷理论等；它也将被用来获得对牛顿引力的汤川型修正、以及暗物质与暗能量假想粒子组分更强的限制。

我们希望本期专刊《卡西米尔效应 75 年：进展与展望》所发表的论文，能推动这一前景广阔的物理学领域进一步发展。

作者贡献、资助与利益冲突

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